

LESSON 318 (2.0 GROUND)

Lesson Objective

During this ground lesson, the student will be introduced to approach charts, instrument approach procedures, and IFR clearances. Other flight publications, including the Aeronautical Information Manual, will also be introduced.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

- Instrument Flying Handbook. ([FAA-H-8083-15B](#))
 - Chapter 10.
- Aeronautical Chart User's Guide. ([Guide](#))
- Terminal Procedures Publication. ([TPP](#)) - SE3.
- Aeronautical Information Manual.
- Flight Operations Manual. ([FOM](#))
 - Part 2: Section 2.0.2.

Lesson Content

Flight Publications - Terminal Procedures Publication

- About:
 - Published every 56 days.
 - Different volumes depending on the region (just like the chart supplement).
 - Revisions:
 - Every 56 days.
 - Additionally, a change notice is published at the midpoint between revisions.
- Structure:
 - Inoperative Components or Visual Aids Table
 - This section indicates how landing minimums must be raised if a component of the approach system is inoperative.
 - Explanation of Terms/Landing Minima Data
 - This is usually located under page B1 in the TPP (refer to the table of contents).
 - IFR Landing Minima
 - Landing minima are established for five aircraft approach categories. The student should memorize these values.

Approach Category	A	B	C	D	E
Speed (Knots)	0-90	91-120	121-140	141-165	Abv 165

- A diagram is provided showing each of the minima values.

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- Potential WAAS outages:
 - The **W** symbol indicates outages of the WAAS vertical guidance may occur daily at this location due to initial system limitations. WAAS NOTAMs for vertical outages are not provided for this approach. Use LNAV minima for flight planning at these locations.
- Cold temperature airports:
 - A **❄**-12°C symbol indicates a cold temperature altitude correction is required at this airport when the reported temperature is at or below the temperature listed.
- Circling approach obstacle protected airspace:
 - A **C** symbol indicates that the approach uses the **expanded** circling approach maneuvering airspace radius. This means the distance we begin our circle at is different for these approaches than those without this symbol. Use the tables found in this section to determine the appropriate distance for each approach.
 - Circling approaches will be discussed more in depth down below.
- Comparable values of RVR and visibility
 - This table converts RVR to visibility. This may be needed in the event of an inoperative RVR system.
- Radar minima
 - This section explains the various values found in the Radar Minimums section (PAR, ASR, etc).
- General Information
 - This is usually located under page J1 in the TPP (refer to the table of contents).
 - This contains information about the TPP itself such as:
 - Issuance times.
 - Information included.
 - Chart currency information.
 - Miscellaneous.
 - Standard terminal arrivals and departure procedures.
 - Procedure PBN/equipment requirements.
 - Pilot controlled airport lighting systems.
- Abbreviations
 - This is usually located under page D1 in the TPP (refer to the table of contents).
 - This section contains a list of various abbreviations and its meanings.
- Legend
 - This section describes the meaning of common symbology found on DPs, STARs, Approach plates, etc.
 - Go through the legend with the student.
 - More information can be found in the Aeronautical Chart User's Guide (TPP).

- Frequency Pairing Table
 - This is usually located under page J1 in the TPP (refer to the table of contents).
 - The frequency channels of the VOR and the TACAN at each VORTAC facility are “paired” in accordance with a national plan to simplify airborne operation.
- Index of Terminal Charts and Minimums
 - This is usually located under page K1 in the TPP (refer to the table of contents).
 - This section is used to locate STARs, DP, IAPs, and airport diagrams for a given airport.
 - Show your student how to find a particular procedure for a given airport.
- IFR Takeoff Minimums, Departure Procedures, and Diverse Vector Area (Radar Vectors)



 - This is usually located under page L1 in the TPP (refer to the table of contents).
 - This section contains all the information about an airport’s departure so that pilots may depart an airport IFR with guaranteed obstacle clearance.
 - These procedures will be further explained in the departure and arrival procedures below.
- IFR Alternate Airport Minimums



 - This is usually located under page M1 in the TPP (refer to the table of contents).
 -  designation on the approach chart indicates that the approach procedure has **non-standard alternate minimums** or restrictions for its use as an alternate. Refer to this section.
 - Alternates will be discussed in more detail in lesson 325.
- Radar Minimums
 - This is usually located under page N1 in the TPP (refer to the table of contents).
 - The only airborne radio equipment necessary for radar approaches is a functional radio transmitter and receiver.

JACKSONVILLE NAS (TOWERS FLD) (KNIP), Jacksonville, FL						
Amdt 7 07SEP23 (24193) (USN)						
RADAR ¹ - (E) 127.7x 266.8x 276.4x 282.375 299.6 328.4 348.0x 363.0x 379.225x 						
					HAT/ HATh/ HAA	ELEV 23
PAR ¹	RWY	GS/TCH/RPI	CAT	DH/ MDA-VIS		CEIL-VIS
	10 ²⁻³	3.0°/40/793	ABCDE	121-¾	100	(100-¾)
	28 ²⁻⁴⁵	3.0°/40/751	ABCDE	113-¾	100	(100-¾)
PAR W/O GS ¹	10 ⁶		ABCDE	400-5/8	379	(400-5/8)
	28 ⁶⁻⁷		ABCDE	340-¾	327	(400-¾)
ASR	28 ⁸⁻⁹		ABCDE	360-¾	347	(400-¾)
	10 ⁸⁻¹⁰		AB	440-½	419	(500-½)
			CDE	440-¾	419	(500-¾)
 CIR	All Rwy		AB	500-1½	477	(500-1½)
			C	560-1½	537	(600-1½)
			D	780-2½	757	(800-2½)
			E	920-3	897	(900-3)

- Land and Hold-Short Operations (LAHSO)
 - This is usually located under page Q1 in the TPP (refer to the table of contents).
 - LAHSO is an acronym for "Land and Hold-Short Operations." These operations

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include landing and holding short of an intersection runway, an intersecting taxiway, or other predetermined points on the runway other than a runway or taxiway. Measured distance represents the available landing distance on the landing runway, in feet.

CITY/AIRPORT	LDG RWY	HOLD-SHORT POINT	AVBL LDG DIST
DAYTONA BEACH, FL			
DAYTONA BEACH INTL (DAB)	07L	TWY W	7,500 feet
	16	07L-25R	2,900 feet
	34	07L-25R	2,564 feet

○ Hot Spots

- This is usually located under page P1/P2 in the TPP (refer to the table of contents).
- A "hot spot" is a runway safety related problem area on an airport that presents increased risk during surface operations. Typically, it is a complex or confusing taxiway/taxiway or taxiway/runway intersection.
- The area of increased risk has either a history of or potential for runway incursions or surface incidents, due to a variety of causes, such as but not limited to: airport layout, traffic flow, airport marking, signage and lighting, situational awareness, and training.
- Hot spots are depicted on airport diagrams as open circles or polygons designated as "HS 1", "HS 2", etc. and tabulated in the list below with a brief description of each hot spot.

CITY/AIRPORT	HOT SPOT	DESCRIPTION*
FORT LAUDERDALE, FL		
FORT LAUDERDALE EXEC (FXE)	HS 1	Int of Twy B and Rwy 09-27.
	HS 2	Int of Twy A and Rwy 13-31.
	HS 3	Int of Twy C, Twy B and Rwy 13-31.
FORT LAUDERDALE, FL		
FORT LAUDERDALE- HOLLYWOOD INTL (FLL)	HS 1	Twy Q at Rwy 10L-28R.
	HS 2	Taxiway Arrival/Departure Risk. Maintain vigilance when taxiing to Rwy 10L on Twy C. Twy B mistaken for Rwy 10L.

○ Standard Terminal Arrival Charts

- This section contains all of the STARs for the region, listed in alphabetical order.
- This will be further explained under the STARs section below.

○ Terminal Charts

- This section includes approach plates, departure plates, and airport diagrams.
- Sorted by city name alphabetically, then the airport name (not code) for each city. Within each airport, the order is: approach plates, the airport diagram, then departure (SID) plates.
- Show the student how to get the approach plates for a particular airport using the TPP (Index of Terminal Charts and Minimums).
 - Show them using ForeFlight as well.

○ Rate of Climb/Descent Table

- This is located on the last page of the TPP.

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- Show them using ForeFlight as well.
- A table is provided for use in planning and executing climbs or descents under known or approximate ground speed conditions.
- Converts FT/NM (for example, takeoff minimums, departure procedures, etc.) or an angle (for example, on a glideslope) to fpm, which is specific to a ground speed and can be read on our VSI.
- Show the student how to convert 200ft/nm and a 3.0° glideslope angle to fpm.

INSTRUMENT TAKEOFF OR APPROACH PROCEDURE CHARTS RATE OF CLIMB/DESCENT TABLE (ft per min)													
A rate of climb/descent table is provided for use in planning and executing climbs or descents under known or approximate ground speed conditions. It will be especially useful for approaches when the localizer only is used for course guidance. A best speed, power, altitude combination can be programmed which will result in a stable glide rate and altitude favorable for executing a landing if minimums exists upon breakout. Care should always be exercised so that minimum descent altitude and missed approach point are not exceeded.													
ft/NM	%	GROUND SPEED (knots)											ANGLE
		60	90	120	150	180	210	240	270	300	330	360	
152	2.50	150	230	300	380	460	530	610	680	760	840	910	1.43
200	3.29	200	300	400	500	600	700	800	900	1000	1100	1200	1.89
210	3.46	210	320	420	530	630	740	840	950	1050	1160	1260	1.98
220	3.62	220	330	440	550	660	770	880	990	1100	1210	1320	2.07

Departure and Arrival Procedures

- IFR Takeoff Minimums:
 - Standard IFR Takeoff Minimums - 91.175(f):
 - Part 121, 125, 129, or 135:
 - Aircraft with 2 engines or less: 1 statute mile visibility.
 - Aircraft with more than 2 engines: 1/2 statute mile visibility.
 - Helicopters: 1/2 statute mile visibility.
 - No pilot may takeoff under IFR from a civil airport having published obstacle departure procedures (ODPs) for the takeoff runway to be used, unless the pilot uses such ODPs or an alternative procedure or route assigned by air traffic control.
 - PART 91: NO T/O MINS.
 - FIT Takeoff Minimums (FOM 2.0.2):
 - The ceiling and visibility (both reported and forecast) must be sufficient to complete an IFR approach at the departure airport.
 - An alternate airport must be available within a one-hour flight from the departure airport if the weather is less than VFR at that departure airport.
- Departure procedures:
 - Provides a transition from the airport to the enroute structure while ensuring obstruction clearance.

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- Pilots operating under Part 91 are encouraged to file and fly a departure procedure when one is available.
- Criteria required to provide obstruction clearance on a departure procedure (AIM 5-2-9):
 1. Climb at 200ft/nm until told to level off or upon reaching the minimum IFR altitude.
 2. Cross the departure end of the runway (DER) after reaching 35' AGL.
 3. Make the initial turn only after reaching 400' AGL.
- Types of Departure Procedures:
 - Obstacle Departure Procedure (ODP)
 - Printed either textually or graphically.
 - Textual ODPs will be listed under DEPARTURE PROCEDURE in the IFR Takeoff Minimums, Departure Procedures, and Diverse Vector Area (Radar Vectors) section of the TPP.
 - Graphical ODPs will include (OBSTACLE) in the title and will be found in the Terminal Charts section of the TPP.
 - Provides obstruction clearance from the terminal area to the enroute structure.
 - Not mandatory to follow but recommended (e.g. night, IMC, terrain).
 - Does not require an ATC clearance unless a SID or radar vectors has been assigned by ATC.

ST PETERSBURG, FL

ALBERT WHITTED (SPG)

TAKEOFF MINIMUMS AND (OBSTACLE) DEPARTURE PROCEDURES

AMDT 4 20APR23 (23110) (FAA)

TAKEOFF MINIMUMS:

Rwy 18, 300-1.

Rwy 36, 300-1 or std. w/min. climb of 845' per NM to 300.

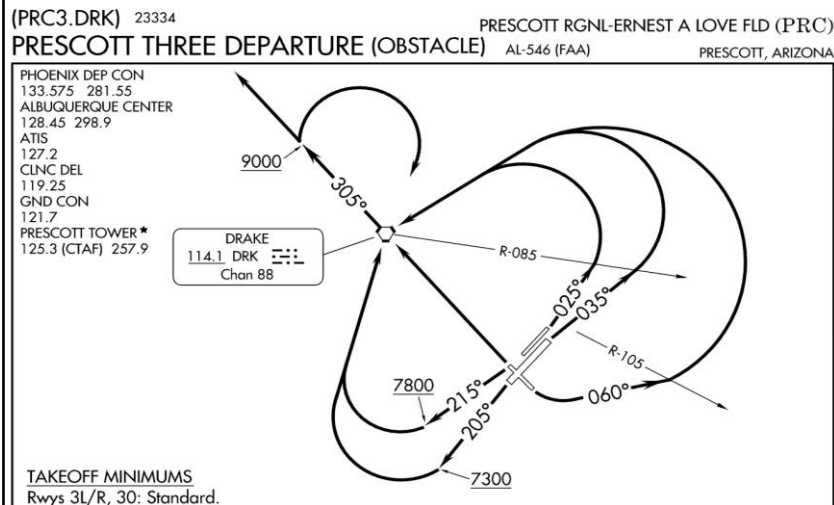
DEPARTURE PROCEDURE:

Rwy 7, climb on heading 067° to 500 before turning left.

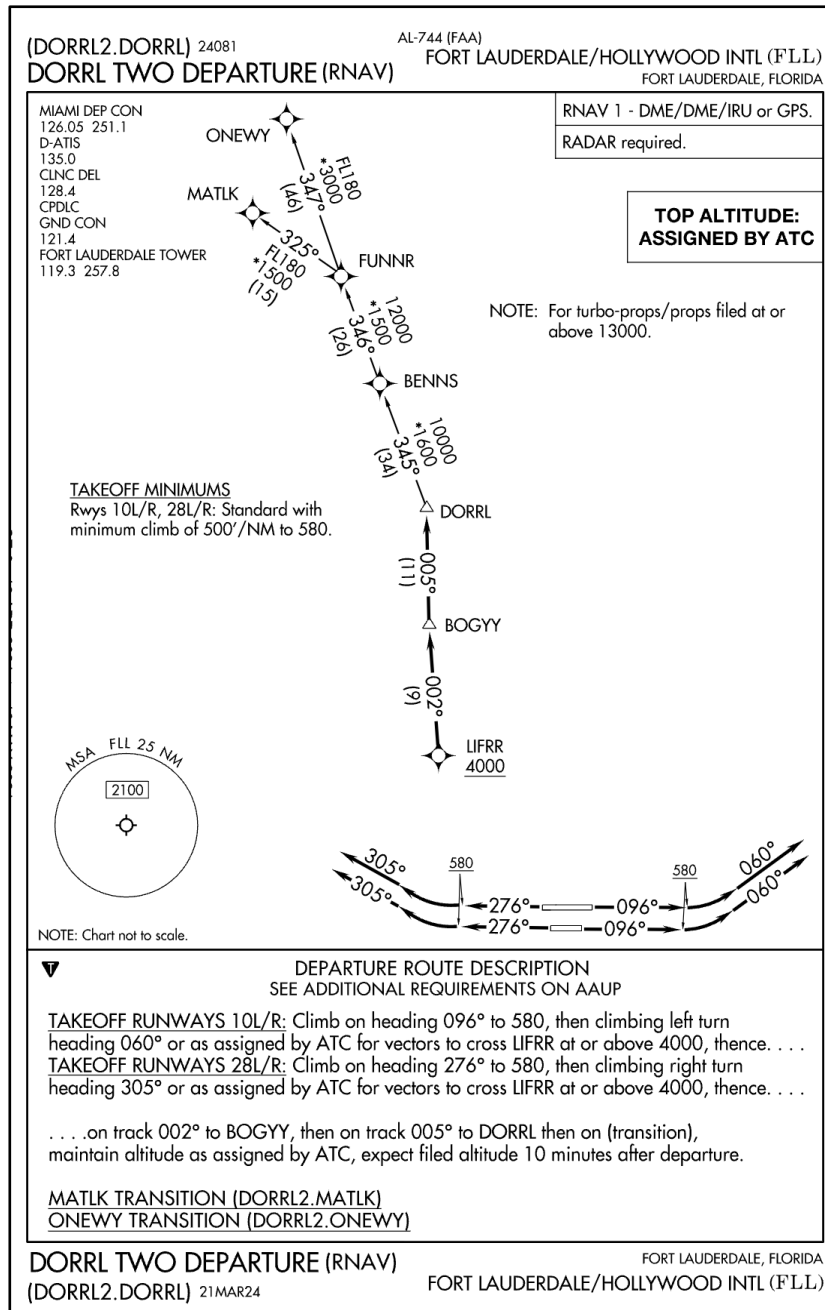
Rwy 18, climb on heading 185° to 500 before turning right.

Rwy 25, climb on heading 247° to 700 before turning right.

Rwy 36, climb on heading 005° to 700 before turning left.



- Standard Instrument Departure (SID)
 - Can be found in the Terminal Charts section of the TPP, just after the approach procedures for a particular airport.
 - Graphical only.
 - Provides obstruction clearance from the terminal area to the enroute structure.
 - Designed to reduce pilot/controller workload.
 - Requires an ATC clearance.
 - Go through the symbology of a SID.
 - Demonstrate how to find and brief a SID.



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■ Diverse Vector Areas

- A Diverse Vector Area (DVA) is an area in which ATC may provide radar vectors until the aircraft reaches a safe altitude.
- Pilots should be aware that Air Traffic facilities may utilize a climb gradient greater than the standard 200 ft/NM within a DVA.
 - See example below for KDVT; upon receiving radar vectors after departure, pilots have to follow the climb gradient (CG) listed in the DVA.



**TAKEOFF MINIMUMS, (OBSTACLE) DEPARTURE PROCEDURES, AND
DIVERSE VECTOR AREA (RADAR VECTORS)**



24081

PHOENIX, AZ (CON'T)
PHOENIX DEER VALLEY (DVT) (CON'T)

Rwy 25L (CON'T), tree, building, antenna, pole beginning 542' from DER, 502' right of centerline, up to 39' AGL/1475' MSL.
Tree, building, pole beginning 630' from DER, 505' left of centerline, up to 1467' MSL.
Pole 1070' from DER, 375' left of centerline, 1468' MSL.
Poles beginning 1077' from DER, 374' left of centerline, up to 39' AGL/1470' MSL.
Pole 1126' from DER, 498' right of centerline, 44' AGL/1478' MSL.
Poles beginning 1183' from DER, 495' left of centerline, up to 41' AGL/1471' MSL.
Signs beginning 1234' from DER, 356' left of centerline, up to 48' AGL/1478' MSL.
Rwy 25R, lighting 30' from DER, 114' left of centerline, 3' AGL/1456' MSL.
Lighting, general utility beginning 30' from DER, 112' right of centerline, up to 2' AGL/1456' MSL.
General utility 31' from DER, 112' left of centerline, 4' AGL/1457' MSL.
Tree 1139' from DER, 798' right of centerline, 1490' MSL.
Terrain 1.3 NM from DER, 2396' right of centerline, 1686' MSL.
Vegetation, tree beginning 1.3 NM from DER, 2270' right of centerline, up to 1697' MSL.
Pole, traverse way beginning 1.3 NM from DER, 1847' right of centerline, up to 30' AGL/1713' MSL.
Antenna 1.3 NM from DER, 2640' right of centerline, 57' AGL/1732' MSL.
Tower, tree, pole, vegetation beginning 1.3 NM from DER, 2073' right of centerline, up to 66' AGL/1741' MSL.
Antenna, tower, traverse way, tree beginning 1.4 NM from DER, 2749' right of centerline, up to 91' AGL/1769' MSL.
Tree 1.6 NM from DER, 3023' right of centerline, 25' AGL/1700' MSL.

DIVERSE VECTOR AREA (RADAR VECTORS)
ORIG 13NOV14 (14317) (FAA)
Rwys 25L, 25R, heading as assigned by ATC; requires minimum climb of 370' per NM to 4600.

■ Visual Climb Over Airport (VCOA)

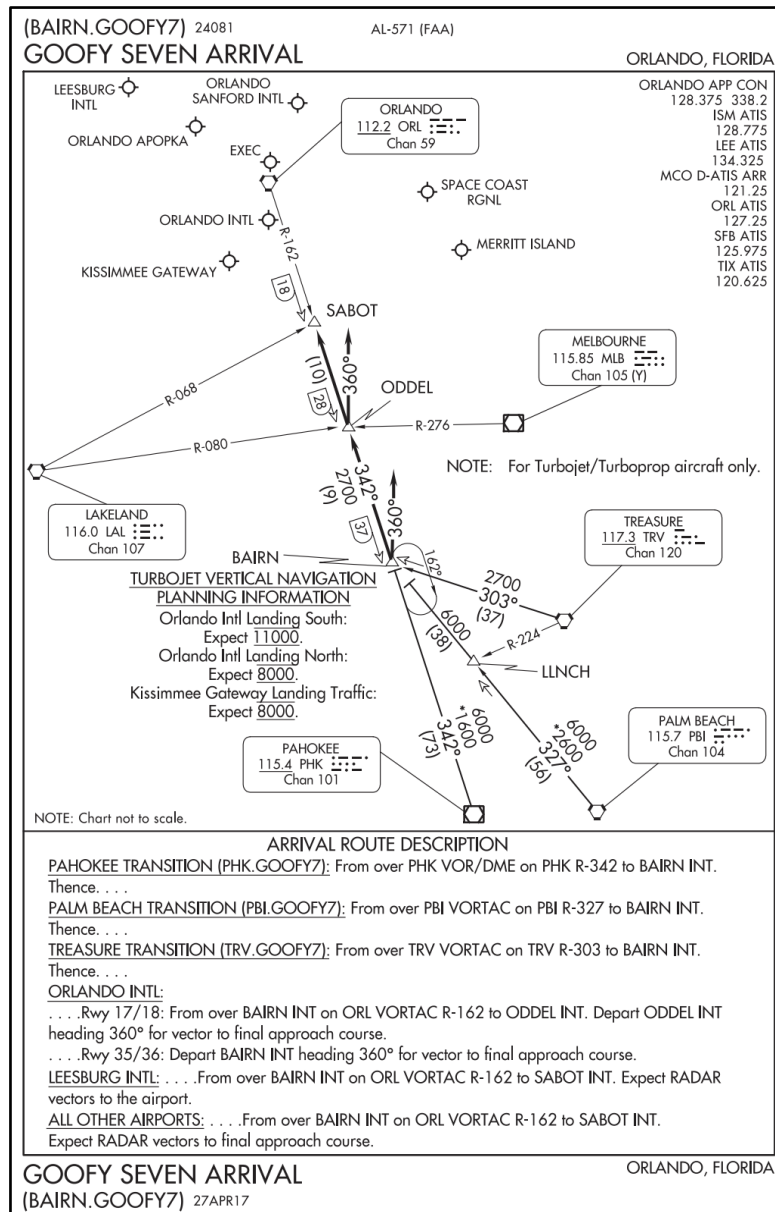
- As the name suggests, a VCOA is a departure option for an IFR aircraft involving a climb while remaining over the field, operating in VMC equal to or greater than the specified visibility and ceiling to altitude where the aircraft can be in IMC and proceed via radar vectors or a fix mentioned in the departure procedure.
- A departure option for an IFR aircraft when there are obstacles 3 miles away from the airport that require a climb gradient of greater than 200 ft/min.

LAKE PLACID, NY
LAKE PLACID (LKP)
TAKEOFF MINIMUMS AND (OBSTACLE) DEPARTURE PROCEDURES
AMDT 1 10OCT19 (19283) (FAA)
TAKEOFF MINIMUMS:
Rwy 14, NA-Obstacles.
Rwy 32, std. w/min. climb of 485' per NM to 4000, or 3600-3 for VCOA.
DEPARTURE PROCEDURE:
Rwy 32, climbing left turn to heading 304° to 5000 before proceeding on course.
VCOA:
Rwy 32, obtain ATC approval for VCOA when requesting IFR clearance. Climb in visual conditions to cross Lake Placid airport at or above 5200 before proceeding on course.

○ Takeoff Obstacle Notes:

- Low close-in obstacles are those obstacles that are located within 1 NM of the DER and penetrate the 40:1 Obstacle Clearance Surface (OCS).
- Low close-in obstacles are noted in the IFR Takeoff Minimums, Departure Procedures, and Diverse Vector Area (Radar Vectors) section .

- Arrival procedures:
 - Standard Terminal Arrival Routes (STARs):
 - These arrival routes simplify clearance delivery procedures for arriving aircraft, specifically for airports that have a high density of traffic.
 - STARs are located towards the front of the TPP in the Standard Terminal Arrival Charts section.
 - Textual and graphical.
 - A pilot may be issued a clearance containing a STAR.
 - To accept this clearance, a pilot must at least have the textual description.
 - It is the pilot's responsibility to accept or deny the issuance of a STAR.
 - A pilot may file an IFR flight plan and put "NO STAR" in the remarks.
 - If a STAR is accepted, it requires mandatory compliance.
 - Demonstrate to the student how to find and brief a STAR.



Approaches

- Types of approaches:
 - Precision: a standard instrument approach procedure in which an electronic glideslope or other type of glidepath is provided.
 - ILS
 - GLS (review GBAS from Stage 1).
 - PAR: The controller issues guidance instructions to the pilot based on the aircraft's position in relation to the final approach course (azimuth), the glidepath (elevation), and the distance (range) from the touchdown point on the runway as displayed on the radar scope.

JACKSONVILLE NAS (TOWERS FLD) (KNIP), Jacksonville, FL

Amdt 7 07SEP23 (24193) (USN)

ELEV 23

RADAR¹ - (E) 127.7x 266.8x 276.4x 282.375 299.6 328.4 348.0x 363.0x 379.225x


	<u>RWY</u>	<u>GS/TCH/RPI</u>	<u>CAT</u>	<u>DH/</u> <u>MDA-VIS</u>	<u>HAT/</u> <u>HATh/</u> <u>HAA</u>	<u>CEIL-VIS</u>
PAR ¹	10 ²³ 28 ²⁴⁵	3.0°/40/793 3.0°/40/751	ABCDE ABCDE	121-¼ 113-¼	100 100	(100-¼) (100-¼)
PAR W/O GS ¹	10 ⁶ 28 ⁶⁷		ABCDE ABCDE	400-⅝ 340-¾	379 327	(400-⅝) (400-¾)
ASR	28 ⁸⁹ 10 ⁸¹⁰		ABCDE AB CDE	360-¾ 440-½ 440-¾	347 419 419	(400-¾) (500-½) (500-¾)
 CIR	All Rwy		AB C D E	500-1⅝ 560-1½ 780-2½ 920-3	477 537 757 897	(500-1⅝) (600-1½) (800-2½) (900-3)

- Non-precision: a standard instrument approach procedure in which no electronic glideslope is provided.
 - VOR
 - TACAN
 - LNAV
 - NDB
 - LOC
 - ASR
 - LDA
 - SDF
- Approaches with vertical guidance: A term used to describe RNAV approach procedures that provide lateral and vertical guidance but do not meet the requirements to be considered a precision approach.
 - LPV: Localizer Performance with Vertical guidance - the CDI needle becomes more sensitive the closer to the runway, like a localizer. Requires WAAS.
 - LNAV/VNAV: Lateral and Vertical Navigation - the CDI needle does not increase in sensitivity. Requires WAAS.

- Contact approach: an approach wherein an aircraft on an IFR flight plan, having an air traffic control authorization, operating clear of clouds with at least 1 mile flight visibility and a reasonable expectation of continuing to the destination airport in those conditions, may deviate from the instrument approach procedure and proceed to the destination airport by visual reference to the surface.
 - This approach will only be authorized when requested by the pilot and the reported ground visibility at the destination airport is at least 1 statute mile.
- Visual approach: An approach conducted on an instrument flight rules (IFR) flight plan which authorizes the pilot to proceed visually and clear of clouds to the airport.
 - The pilot must, at all times, have either the airport or the preceding aircraft in sight.
 - This approach must be authorized and under the control of the appropriate air traffic control facility.
 - Reported weather at the airport must be ceiling at or above 1,000 feet, and visibility of 3 miles or greater.
- No-gyro approach: A radar approach/vector provided in case of a malfunctioning gyro-compass or directional gyro.
 - Instead of providing the pilot with headings to be flown, the controller observes the radar track and issues control instructions “turn right/left” or “stop turn” as appropriate.

Approach plate structure:

- Briefing strip - contains general information about the approach.
 - Approach name
 - Airport and airport code
 - Approach frequency
 - Final approach course (note that this may not always be what we twist the CDI to, such as on a LOC BC)
 - Runway landing distance (Rwy ldg)
 - Touchdown zone elevation (TDZE)
 - Airport elevation (Apt Elev)
 - General notes about the approach, such as equipment requirements or anything that is nonstandard (includes takeoff notes and nonstandard alternate minimums).
 - Approach lighting: depicted as a name, code, and diagram. Refer to the legend (around page I2) for further information.
 - Missed approach instructions: textual description
 - Airport and approach control frequencies
- Plan view - a top-down view which mainly depicts the lateral track of the approach.
 - Final approach course
 - Navaid track (localizer/radial/etc.)
 - IAF/IF/FAF locations and means of identification

- Missed approach instructions: graphical view
- MSA: Minimum Safety Altitude
 - Provides at least 1,000 feet of obstacle clearance within a 25-mile radius of the navigation facility, waypoint, or airport reference point upon which the MSA is predicated. MSAs are for emergency use only and do not necessarily assure acceptable navigational signal coverage.
- Profile view - a “side” view which mainly depicts the vertical track of the approach.
 - IAF/IF/FAF of the approach and their distances to the runway and/or navaid
 - Stepdown fixes
 - A fix permitting additional descent within a segment of an instrument approach procedure by identifying a point at which a controlling obstacle has been safely overflown.
 - Depicted as solid lines above and/or below crossing altitudes. 1600
 - Glideslope depiction
 - Glideslope angle and Threshold Crossing Height (TCH)
 - Glideslope intercept point and altitude
 - VDP (explained below)
 - Missed approach point
 - Missed approach instructions: graphical overview
 - Minimums (explained below)
- Airport diagram
 - Airport elevation
 - Touchdown zone elevation
 - Final approach course is depicted as an arrow. Useful for determining direction to turn when circling.
 - Approach lighting
 - Airport lighting
 - Runway layout, lengths, and widths
 - **D** symbol indicates Runway Declared Distance information is found in the Chart Supplement.
- Approach segments
 - Feeder routes
 - A route depicted on IAP charts to designate routes for aircraft to proceed from the en route structure to the IAF.
 - Depicted as thin lines.
 - DME arcs
 - Some approaches include DME arcs as part of the approach procedure itself.
 - The aircraft may need to descend during the arc (ex: VOR Rwy 15 @ KMTN).
 - Course reversal
 - Some approach procedures do not permit straight-in approaches unless pilots are

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being radar vectored. In these situations, pilots are required to complete a procedure turn (PT) or other course reversal, generally within 10 NM of the PT fix, to establish the aircraft inbound on the intermediate or final approach segment.

- When is a procedure turn not required? (SHARPTT)
 - Straight-in approach clearance
 - Holding in lieu of a procedure turn
 - DME Arc
 - Radar vectors to final
 - No PT depicted on chart
 - Timed approach from a hold fix
 - Teardrop course reversal
- Initial approach segment
 - The segment between the initial approach fix (IAF) and the intermediate fix (IF) or the point where the aircraft is established on the intermediate course or final approach course.
- Intermediate approach segment
 - The segment between the intermediate fix or point (IF) and the final approach fix (FAF).
- Final approach segment
 - The segment between the final approach fix or point (FAF) and the runway, airport, or missed approach point (MAP).
- Missed approach segment
 - The segment between the missed approach point (MAP) or the point of arrival at decision height and the missed approach fix at the prescribed altitude.

VDP

- Identified by the symbol () , is a defined point on the final approach course of a non-precision straight-in approach procedure from which a stabilized visual descent from the MDA to the runway touchdown point may be commenced. The VDP will be identified by DME or RNAV along-track distance to the MAP. (AIM 5-4-5 h)
- The pilot should not descend below the MDA prior to reaching the VDP. The VDP distance is based on the lowest MDA published on the IAP and harmonized with the angle of the visual glide slope indicator (VGSI) (if installed) or the procedure VDA (if no VGSI is installed).
- A VDP may not be published under certain circumstances which may result in a destabilized descent between the MDA and the runway touchdown point.
 - To calculate VDP when not published = $MDA (AGL)/300$, this ensures a three-degree slope to the runway
- If a FAF (depicted by the Maltese cross) does not exist, final descent can be planned to use the same calculation: $Altitude\ to\ lose/300 = distance\ in\ NM\ before\ a\ given\ point\ to\ begin\ descent$.

Minimums

- DA: a specified altitude in the precision approach at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.
- MDA: the lowest altitude, expressed in feet MSL, to which descent is authorized on final approach or during circle-to-land maneuvering in execution of a standard instrument approach procedure (SIAP) where no electronic glideslope is provided.
- Show and explain to the student the values found in the landing minima section of the approach plate:

LANDING MINIMA FORMAT

In this example airport elevation is 1179, and runway touchdown zone elevation is 1152.

	DA	Visibility (RVR 100's of feet)	Aircraft Approach Category	HAT
CATEGORY	A	B	C	D
Straight-in ILS to Runway 27 → S-ILS 27	1352/24		200	(200-½)
Straight-in with Glide Slope Inoperative or not used to Runway 27 → S-LOC 27	1440/24	288	(300-½)	1440/50 288 (300-1)
CIRCLING →	1540-1 361 (400-1)	1640-1 461 (500-1)	1640-1½ 461 (500-1½)	1740-2 561 (600-2)
	MDA	HAA	Visibility in Statute Miles	

All weather minimums in parentheses not applicable to Civil Pilots.
Military Pilots refer to appropriate regulations.

- 91.175(c): Operation below DA/DH or MDA
Emphasize to the student the importance of memorizing this regulation.
 1. The aircraft is continuously in a position from which a descent to a landing on the intended runway can be made at a normal rate of descent using normal maneuvers, and for operations conducted under part 121 or part 135 unless that descent rate will allow touchdown to occur within the touchdown zone of the runway of intended landing;
 2. The flight visibility is not less than the visibility prescribed in the standard instrument approach being used;
 3. Except for a Category II or Category III approach where any necessary visual reference requirements are specified by the Administrator, at least one of the following visual references for the intended runway is distinctly visible and identifiable to the pilot:
 - The approach light system, except that the pilot may not descend below 100 feet above the touchdown zone elevation using the approach lights as a reference unless the red terminating bars or the red side row bars are also distinctly visible and identifiable.
 - The threshold.
 - The threshold markings.
 - The threshold lights.
 - The runway end identifier lights.
 - The visual glideslope indicator.
 - The touchdown zone or touchdown zone markings.
 - The touchdown zone lights.
 - The runway or runway markings.
 - The runway lights.

Missed approach

- A maneuver conducted by a pilot when an instrument approach cannot be completed to a landing. The route of flight and altitude are shown on instrument approach procedure charts. A pilot executing a missed approach prior to the Missed Approach Point (MAP) must continue along the final approach to the MAP.
- At locations where ATC radar service is provided, the pilot should conform to radar vectors when provided by ATC in lieu of the published missed approach procedure.
- If we are using a glideslope, execute the missed approach upon reaching the DA.
- If we are not using a glideslope, hold the Minimum Descent Altitude (MDA) until reaching the Missed Approach Point (MAP).
- The Missed Approach Point (MAP) may not be published as a part of the approach (ex: ILS 36 @ KTIK). When this occurs, use the “FAF to MAP” box shown at the bottom of the airport diagram on the approach plate to calculate the MAP.
- Show the student examples of several different approaches and explain when to go missed.

Circling approach

- A maneuver initiated by the pilot to align the aircraft with a runway for landing when a straight-in landing from an instrument approach is not possible or is not desirable. (Ex: weather, runway length, lighting, etc.)
- At tower-controlled airports, this maneuver is made only after ATC authorization has been obtained and the pilot has established required visual reference to the airport.
- Review the circling approach obstacle protected airspace  tables found in the Terms/Landing Minima Data section of the TPP.
- Approaches that end in letters at the beginning of the alphabet (RNAV-A, LOC/DME-E, etc.) are circling-only. Circling-only approaches are published for the following criteria:
 1. The final approach course alignment with the runway centerline exceeds 30°.
 2. The descent gradient is greater than 400 ft/NM from the FAF to the threshold crossing height (TCH).
 3. A runway is not clearly defined on the airfield.

IFR Clearances

- When is a clearance required?
 - According to 14 CFR 91.173, no person may operate an aircraft under IFR in controlled airspace unless an IFR flight plan has been filed and a clearance has been received from ATC.
 - Although flying into IMC in uncontrolled airspace without a clearance is technically legal, it has been found in the past to be a violation of **91.13 - Careless or reckless operation**.
- Obtaining an IFR clearance
 - Towered airport: Contact clearance delivery or ground frequency. The ATIS will sometimes specify a frequency.

- Non-towered airport: Consult chart supplement / AFD for frequency or ARTCC phone number.
 - Depart VFR and contact ATC.
 - Contact ATC on the ground. Expect a *Clearance Void Time* or *EDCT (Expect Departure Clearance Time)*.
 - Clearance void time: The latest time at which one may depart IFR. A pilot who does not depart prior to the clearance void time must advise ATC as soon as possible of their intentions.
 - Ground communication outlet (GCO): on the ground, click mic 6 times to contact FSS & 4 times to contact ATC.
- Elements of an IFR clearance:
 - **C:** Clearance Limit
 - **R:** Route
 - **A:** Altitude
 - **F:** Frequency
 - **T:** Transponder

TECH FLIGHT 645, cleared to the Sarasota International Airport as filed, after departure maintain 2000ft, expect 6000ft 10 minutes after departure, departure frequency is 132.65, squawk 0812.

- Amended IFR clearance
 - Amendments to the initial clearance will be issued at any time an air traffic controller deems such action necessary to avoid possible conflicts between aircraft.
 - Clearances will require that a flight “hold” or change altitude prior to reaching the point where standard separation from other IFR traffic would no longer exist
- Cruise clearance: AIM 4-4-4
 - The term “cruise” may be used instead of “MAINTAIN” to assign a block of airspace to a pilot from the minimum IFR altitude up to and including the altitude specified in the cruise clearance.
 - The pilot may level off at any intermediate altitude within this block of airspace. Climb/descent within the block is to be made at the discretion of the pilot. However, once the pilot starts descent and verbally reports leaving an altitude in the block, the pilot may not return to that altitude.
- Approach clearance:
 - Authorization by ATC for a pilot to conduct an instrument approach. The type of instrument approach for which a clearance and other pertinent information is provided in the approach clearance when required.
 - “Fly heading 060, maintain 2000 until established, cleared for the ILS 09R.”

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 319.
- Assign Homework:
 - Continuously review the contents of this lesson.
 - Review the approach plates for a VOR, LOC, and ILS approach of your choosing.
 - For example, the VOR 09R @ KMLB, LOC 09R @ KMLB, and ILS 09R @ KMLB.
 - Review current NOTAMs and update landing minimums as necessary.

Completion Standards

This lesson shall be complete when through oral quizzing and discussion:

1. The student displays a basic understanding of approach charts, non-precision and precision approaches, and ATC clearances.